

Yongle Duanmu

Homepage — GitHub — Email: HPU_DMYL@163.com

EDUCATION BACKGROUND

Dalian University of Technology

M.S. in Control Science and Engineering

Dalian, China

Sep. 2025 – Jun. 2028

- GPA: **4.37 / 5.00** (Top 1%)
- Research Interests: ECG-based Emotion Recognition, Multimodal Learning, Domain Adaptation, Embedded AI, Edge Computing, Intelligent Healthcare.

Henan Polytechnic University

B.Eng. in Automation, School of Electrical Engineering and Automation

Jiaozuo, China

Sep. 2021 – Jun. 2025

- GPA: **4.21 / 5.00** (Rank 1); ranked **first in the major for four consecutive years**.
- Strong background in control engineering, embedded systems, circuit design, signal processing, programming, and mathematical modeling.

INTERNSHIP EXPERIENCE

Dalian Institute of Chemical Physics, Chinese Academy of Sciences

Embedded Software Development Intern, Battery Thermal Runaway Project

Dalian, China

May 2026 – Aug. 2026

- Developed an implantable multi-source data acquisition platform for early warning of thermal runaway in new-energy vehicle batteries, monitoring strain, temperature, CO/CO₂/H₂/smoke, PM2.5, and other parameters.
- Independently designed a high-integration embedded PCB, including schematic design, layout, hardware bring-up, and signal integrity verification with oscilloscopes and multimeters.
- Built a multi-task real-time acquisition system based on STM32F4 and FreeRTOS, with drivers for SPI, IIC, CAN, USART, SD card storage, and multiple sensing modules.
- Designed a battery-tab signal multiplexing scheme to encode, modulate, transmit, and recover sensing data under space-constrained battery-embedded conditions.

Xiaomi

Embedded Software Development Intern, OpenVela Team

China

Sep. 2026 – Aug. 2027

- Upcoming internship in embedded software development and operating-system-related engineering.

MAIN RESEARCH

1. **Emotion-Aware Sign Language Recognition via Cross-Modal Fusion of ECG Signals and Skeletal Data**
Yongle Duanmu, et al.
ETAI 2026, Best Paper Award. First author.
2. **ECG-MFN: An Attention-Guided Multimodal Fusion Network for ECG-Based Emotion Recognition**
Yongle Duanmu, et al.
Manuscript submitted to Biomedical Signal Processing and Control. First author.
3. **Mechanism-Constrained Gas-Response Decomposition and Sensor-Array Optimization for Lithium Iron Phosphate Battery Vent-Gas Sensing**
Haotian Zhang, Yongle Duanmu, et al.
Manuscript submitted to Sensors and Actuators B: Chemical. Co-first author; internship output.

SELECTED PROJECTS & COMPETITIONS

Bidirectional Intelligent Bionic Sign Language Translator

2024

Team Leader; National First Prize, National Undergraduate Embedded Chip and System Design Competition

- Designed a multimodal sign-language translation system integrating embedded sensing, gesture recognition, signal acquisition, and intelligent human-computer interaction.

Intelligent Medication Intervention System Based on Multimodal AI and Edge Computing

2024

Team Leader; National First Prize, Chinese Collegiate Computing Competition

- Built an intelligent medication system integrating K230 edge vision, i.MX6ULL gateway, local medical LLM, RAG knowledge base, health monitoring, behavior detection, and 3D-printed mechanical structure.

Dynamic Optimization of Insurance Enrollment under Extreme Weather

2024

Team Leader; Meritorious Winner, Mathematical Contest in Modeling

- Developed a mathematical modeling and optimization framework for insurance enrollment strategies under extreme weather conditions.

HONORS & AWARDS

- **National First Prize** (Top 1%), National Undergraduate Embedded Chip and System Design Competition, 2024.
- **National First Prize** (Top 1%), Chinese Collegiate Computing Competition, 2024.
- **Meritorious Winner** (Top 6%), Mathematical Contest in Modeling, 2024.

SKILLS

- **Embedded Systems:** STM32, FreeRTOS, SPI, IIC, CAN, USART, SD card storage, sensor drivers, PCB schematic design and layout, hardware debugging.
- **AI & Software:** Python, C/C++, ECG-based emotion recognition, multimodal learning, domain adaptation, edge AI deployment, intelligent healthcare systems.
- **Tools & Languages:** Oscilloscope, multimeter, SolidWorks, Git; Chinese native, English working proficiency.